MGMTMSA-400-01/02: Homework 2

Due Date: October 25, 2025, 11:59 PM
No Late Homework!

Submission: Submit your write-up on Gradescope and your code or workbooks on Bruin Learn.

Note: This assignment has a total score of 40 points. Your score will be converted to 10 points toward the final grade. You may want to set the seed using `set.seed(123)` in R or `numpy.random.seed(123)` in Python, as in your last homework.

Bonus Points: Question 3(f) is optional and counts only for bonus credit. Your final score will not exceed 40.

Question 1: Album Sales (15 points)

There are two major categories of costs associated with producing music albums: (a) production and recording, and (b) marketing and promotion. As a music industry consultant, Ben wants to understand how these two types of spending affect album sales.

To study this, Ben collects data on 200 randomly selected albums released in recent years. The dataset includes each album's first-year sales (labeled `sales`), total production and recording cost (labeled `prod_cost`), and marketing and promotion cost (labeled `marketing`). Costs are measured in hundreds of thousands of dollars, and sales are measured in thousands of copies.

Because both costs and sales are highly right-skewed, Ben takes logarithms of all three variables before analysis. He begins by summarizing the data and then runs two separate least-squares regressions, each using logged sales as the dependent variable and one of the cost measures as the independent variable.

The `R` output below summarizes the regression results. Some entries are intentionally omitted. You may need to infer them. **Some questions may not be answerable using the provided information. In those cases, please explain what additional information is needed.**

```
> # Compute sample mean and standard deviation for three variables
> c(mean(log(albums$sales)), sd(log(albums$sales)))
[1] 3.778082 2.138198
> c(mean(log(albums$prod_cost)), sd(log(albums$prod_cost)))
[1] 1.516873 1.056563
> c(mean(log(albums$marketing)), sd(log(albums$marketing)))
[1] (--a--) 0.9158577

> # Linear regressions
> model1 <- lm(log(sales) ~ log(prod_cost), data = albums)
```

```
> summary(model1)

Call:
lm(formula = log(sales) ~ log(prod_cost), data = albums)

Residuals:
    Min      1Q  Median      3Q     Max
-5.1281 -1.3931 -0.0548  1.1938  6.4912

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)      2.5897     0.2449  10.573  < 2e-16
log(prod_cost)   0.7835    (--b--) (--c--) 1.49e-08

Residual standard error: 1.976 on 198 degrees of freedom
Multiple R-squared:  0.1499

> model2 <- lm(log(sales) ~ log(marketing), data = albums)
> summary(model2)

Call:
lm(formula = log(sales) ~ log(marketing), data = albums)

Residuals:
    Min      1Q  Median      3Q     Max
-5.2505 -1.4419 -0.2091  1.4196  6.0353

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)      3.4641     0.1542  22.458  < 2e-16
log(marketing)   0.8132     0.1555   5.229 4.32e-07

Residual standard error: (--d--) on 198 degrees of freedom
Multiple R-squared:  0.1213
```

(a) Which variable exhibits a stronger correlation with logged sales: `log(prod_cost)` or `log(marketing)`? If there is not enough information to determine this, explain why. (3 points)

(b) What is the mean of logged marketing costs, i.e., the value of `(--a--)`? (3 points)

(c) What is the standard error of `log(prod_cost)` in regression `model1`, i.e., the value of `(--b--)`? (3 points)

(d) What is the residual standard error for regression `model2`, i.e., the value of `(--d--)`? (3 points)

(e) Ben's manager asks him to express sales in millions of copies instead of thousands by dividing `sales` by 1,000, then rerun `model2` using the rescaled variable. Without rerunning the regression, can you determine the new estimated intercept, estimated slope, and R^2? If so, report these values and explain your reasoning. If not, explain why. (3 points)

Question 2: Conditional Distribution and Prediction (8 Points)

In this question, we will use data from `simu_yx.csv` to understand how input variables may help with the prediction of an output variable y.

You are provided with the dataset `simu_yx.csv`, which contains 1000 observations on three variables: y, x_1, and x_2. y is the output variable and is the variable we plan to predict. x_1 and x_2 may be used as input variables. You may notice that x_1 and x_2 both have sample means of zero.

(a) Randomly select 10, 100, and 1000 observations from the dataset, and run regressions of y on x_1 using the selected observations. Report the regression results. (2 points)
Hint: You can use the code `df_sample <- df[sample(nrow(df), 10), ]` to randomly select 10 observations from the data frame `df`.

(b) Using the regression results from (a), discuss how R^2 and the standard error of the slope change with sample size. (3 points)

(c) Compute the correlation coefficient between x_2 and y, as well as the sample means of both variables. Using only this information, are you able to compute the estimated slope and intercept from a regression of y on x_2? If so, report these values. If not, explain why. (3 points)

Question 3: Rider Demand for a Ride-Hailing App (17 Points)

In this question, we investigate data from `Rider-Request.RData`. This synthetic dataset documents riders' request sessions on a ride-hailing app in Houston, TX, for three weeks. Each observation corresponds to a rider request session, i.e., a rider opens the app, selects a pickup location and destination, and observes the trip fare and estimated driver arrival time. The session may result in a trip request (`request` = 1) or not (`request` = 0).

Below is a data dictionary describing all variables in the dataset:

- `origin_hex`: H3 address of the rider's location when the trip was requested or when the session ended.
- `origin_lat`: Latitude of the rider's location when the trip was requested or when the session ended.
- `origin_lng`: Longitude of the rider's location when the trip was requested or when the session ended.
- `home_dist`: Distance (in miles) between the rider's location and their home.
- `request`: Binary variable equal to 1 if the trip was requested and 0 otherwise.

- `dest_hex`: H3 address of the trip destination.
- `dest_lat`: Latitude of the destination.
- `dest_lng`: Longitude of the destination.
- `multiplier`: Surge multiplier applied to the trip fare.
- `income`: Rider's income, approximated using the median household income of the ZIP code associated with the rider's home address.
- `airport`: Binary variable equal to 1 if the destination is George Bush Intercontinental Airport and 0 otherwise.
- `base_fare`: Base fare for the trip before surge pricing.
- `eta`: Displayed pickup time (in minutes) shown to the rider.
- `arrival_time`: Local time when the rider opened the app (i.e., when the session started).
- `trip_distance`: Straight-line (great-circle) distance between the origin and destination.

In particular, base fares consist of two components: a fixed booking fee ($2.30) applied to all requests, and a variable fare that depends on the request time, trip distance, and the estimated duration of the trip. The surge multiplier applies only to the variable portion of the fare. This is when the app indicates that "fares are high" or "very high." For example, consider a trip with a base fare of $9.30 and a multiplier of 1.3. The total fare shown to the rider would be calculated as $\$2.30 + (\$9.30 - \$2.30) \times 1.3 = \$11.40$.

In the following questions, we are going to use estimated pickup time (`eta`) as an input variable to predict total trip fares.

(a) Compute the total trip fare for every session using the base fare and surge multiplier. Plot separate histograms of total trip fares and estimated pickup times, and briefly describe the shape of each distribution. (3 points)

(b) Estimate a simple linear regression that captures the relationship between the two variables from part (a), using the logged trip fare as the dependent variable and the non-transformed pickup time as the independent variable. From the regression output, report the estimated slope and intercept along with their standard errors. (3 points)

(c) Interpret the economic meaning of the estimated slope from (b). (2 points)

(d) Construct a 90% confidence interval for the estimated slope. (3 points)

(e) Now restrict your sample to all app sessions from 10:00 AM to 4:00 PM on weekdays (Monday through Friday). Using these sessions, run the same regression as in (b) and compute the 90% confidence interval for the slope. (4 points)
Hint: You can use `as.POSIXlt()` from base R (no additional package needed) to access the hour and weekday of `arrival_time`. Set `tz = "America/Chicago"` to use Houston local time. You could use the following R code:

```
local_time <- as.POSIXlt(dest$arrival_time, tz = "America/Chicago")
is_daytime <- local_time$wday %in% 1:5 & local_time$hour >= 10 & local_time$hour < 16
daytime_sessions <- dest[is_daytime, ]
```

(f) Compare the estimated slopes from (b) and (e). Which one is larger? Explain why. (bonus points: 2)

(g) Compare the widths of the confidence intervals from (d) and (e). Which one is wider? Explain why. (2 points)